

Dead forests burning: the influence of beetle outbreaks on fire severity and legacy structure in sub-boreal forests

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Abstract. Recent regional mountain pine beetle (MPB) outbreaks have generated unprecedented tree mortality across the fire-prone landscapes of western North American forests and could potentially modify fire severity and postfire ecological effects. In 2012, 2013, and 2014, three fires burned through high mortality, gray-phase lodgepole pine-dominated forests in the plateau regions of central interior British Columbia, Canada, providing an opportunity to test for interactions between MPB outbreaks and wildfires. We inventoried 63 plots that spanned gradients of outbreak severity, fire severity, and burning conditions in a wilderness setting. Our objective was to evaluate the influence of outbreak severity on fire severity by assessing typical first-order fire effects as well as legacy structure related to the consumption of woody biomass on snags/trees. We found no evidence of a relationship between outbreak severity and fire severity for six of seven first-order fire effects, with the exception of deep charring. We found evidence that legacy structure in the form of consumed branch structure and deep char development had greater odds of occurrence on MPB-killed snags compared to trees killed during wildfire. Our results indicate two key findings. First, fire severity as it relates to most first-order fire effects measures is not influenced by outbreak severity, instead it is more strongly influenced by the interaction of fuels, weather, and topography during fire events. Second, our results highlight how the interaction between outbreak severity and fire severity alters postfire structural legacies and their functional attributes, which could have important ecosystem implications.

Key words: deep char; *Dendroctonus ponderosae*; legacy structure; mountain pine beetle; *Pinus contorta* var. *latifolia*; short-interval disturbances; snags; sub-boreal; wildfire.

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INTRODUCTION

Forest ecosystems across western North America are increasingly experiencing ecological disturbances from wildfires burning through landscapes with abundant tree mortality from insect outbreaks. The recent mountain pine beetle (*Dendroctonus ponderosae*, hereafter MPB) outbreaks circa the 1990s and 2000s are responsible for tree mortality in forests that span over 25 million hectares across the western United States and Canada (Raffa et al. 2008, Bentz et al. 2010, Meddens et al. 2012), and British Columbia (BC) houses nearly 20 million of those hectares

(Axelson et al. 2009, Perrakis et al. 2014). The spatial extent and high mortality rates associated with recent outbreaks alter standing woody fuels in affected forests from mostly alive to mostly dead, which changes the composition of the fuel profile and raises concerns for increased fire severity (Hicke et al. 2012, Jenkins et al. 2012). The overlap between MPB outbreak and wildfire disturbances that recur within a short time interval may lead to linked effects in which the first event alters the extent, severity, or probability of occurrence for the second event (Kulakowski and Veblen 2007, Simard et al. 2011). Previous field-based studies have investigated interactions of

short-interval MPB-fire disturbances with variable levels of lodgepole pine (*Pinus contorta* var. *latifolia*) mortality in montane regions of the western United States (Harvey et al. 2014a, b, Agne et al. 2016) and found fire severity to be either weakly linked or unrelated to outbreak severity. However, the magnitude of the MPB outbreaks in BC far exceeds the conditions seen in the western United States (Raffa et al. 2008) and the biophysical environment differs from earlier studies (Harvey et al. 2014b, Agne et al. 2016), such that further investigation is required to understand the implications of fire burning through BC's MPB-affected forests.

The changes in fuel profiles from MPB outbreaks and subsequent stand breakdown have raised concerns among land managers for altered fire behavior and potential changes in subsequent fire severity that could generate burning conditions that are more hazardous and more severe than from fire burning through stands of live trees. Severe tree mortality alters the configuration, continuity, and moisture content of fuels over time as stands break down (i.e., needle loss, branch breakage, sloughing bark, snag fall), all of which may influence fire behavior (Page and Jenkins 2007, Hicke et al. 2012). Dry, dead fuels ignite more quickly (Stockstad 1979) and at lower temperatures (Stockstad 1975) compared to live fuels. When these dead fuels are coarse, they are prone to smoldering (Brown et al. 2003), which often extends burning time beyond the initial flaming front (Alexander 1982), thus allowing for dry dead fuels to burn longer and have more biomass consumed (Brown et al. 2003, Hyde et al. 2011) that could alter the structural legacies that persist postfire. The only known empirical study examining the effects of altered fuel profiles on fire behavior found that spread rates increased through red-phase outbreak conditions in lodgepole pine forests in BC (Perrakis et al. 2014). Simulation models posit crown fire to increase during the red phase of the outbreaks, 1–3 years postattack, and then decline as needles are dropped from the canopy and snags transition to the gray-phase of the outbreaks, 3–10 years postattack (Hicke et al. 2012). Alternative models suggest a shift from active crown fire during the red phase to passive crown fire in the gray phase (Klutsch et al. 2011, Simard et al. 2011, Schoennagel et al. 2012) that could result in

more biomass consumption and simplification of the legacy structure of snags.

Retrospective data that evaluate fire effects to characterize fire severity provide a complement to measures of fire behavior for understanding interactions between MPB outbreak and wildfire. Fire severity is often characterized by measurements of first-order fire effects (Reinhardt et al. 2001, Ryan and Elliot 2005) and refers to the amount of immediate ecological change associated with vegetation mortality and biomass loss from fire (Keeley 2009). Retrospective studies on MPB-fire interactions are two pronged either using remotely sensed data (e.g., satellite imagery) to quantify the amount of change between prefire and postfire conditions at coarser resolutions, or field studies that measure fire effects on the ground to characterize fire severity at finer scale resolutions. Existing remote sensing studies have shown that outbreak severity does not increase fire likelihood (Meigs et al. 2015), fire severity (Meigs et al. 2016), or area burned (Hart et al. 2015) for forests in the western United States.

Field studies can capture subtleties that may be absent in remote sensing studies and have found that the relationship between outbreak severity and fire severity varies across the western United States. These studies have focused on subalpine lodgepole pine forests (Harvey et al. 2014a, Agne et al. 2016) and forests dominated by subalpine fir (*Abies lasiocarpa*) but with substantial basal area of lodgepole pine (Harvey et al. 2014b) across topographically complex landscapes. Generally, these studies have suggested that gray-phase outbreak severity results in decreased fire severity (Harvey et al. 2014a, Agne et al. 2016), or limited to no change in fire severity (Harvey et al. 2014b, Agne et al. 2016)—with the exception of deep char, a metric of fire severity that showed a consistently positive relationship with severity of MPB outbreaks (Harvey et al. 2014b). Some measures of fire severity increased under extreme fire weather and were attributed to burning conditions, including deep char (Harvey et al. 2014b), suggesting that prefire beetle outbreak and burning conditions contribute to deep charring on wood. Deep char is generated through incomplete combustion of deadwood often from long, smoldering burns (Bird et al. 2015) that result in more biomass

consumption and is visually distinct compared to scorch that is generated from flaming combustion and typically occurs on trees that are alive at the time of fire (Campbell et al. 2007). Deep char is distinguished by its iridescent black with patterning like the scales of alligator skin in contrast to the matte black, dusty appearance of scorch. Deep charring on trees changes the structure and function of the postfire landscape (Campbell et al. 2007, Donato et al. 2016) by altering structural legacies and has been clearly recognized as an important severity metric when examined in areas of high-severity reburns (fire + fire; Donato et al. 2016). However, the deep char effect, and the altered structural legacy it contributes to the postfire landscape, has largely been ignored in the context of insect outbreak and wildfire interactions.

Here, we examine the effect of gray-phase outbreak severity on fire severity for lodgepole pine-dominated forests with high prefire mortality rates, in central interior BC. Our objective was to evaluate the influence of outbreak severity on fire severity by assessing first-order fire effects after three recent wildfires that burned in 2012, 2013, and 2014. We wanted to (1) ascertain whether the extensive MPB-induced tree mortality that spans the sub-boreal forests of BC responds similarly in terms of first-order fire effects to forests that have burned and been studied in the western conterminous United States and (2) expand understanding and recognition of how postfire legacies (e.g., snags) can be affected by MPB outbreaks. We anticipated first-order fire effects (e.g., scorch/char height and area on trees, surface char, exposed mineral soil) would be unaffected by the severity of the outbreaks and primarily driven by fire weather, based on previous findings (Harvey et al. 2014a, b, Agne et al. 2016). In the context of structural legacies, we anticipated that snags killed by the MPB outbreak a decade prior to fire would burn longer, through smoldering combustion that would consume more wood biomass and lead to consistent development of deep char. We also predicted the interaction between outbreak severity and fire would reduce the structural complexity on snags, due to the potential extended duration of smoldering combustion in addition to prefire stand breakdown where MPB-killed trees experience needle loss, branch breakage, and shedding of bark. In

contrast, the legacies of trees that were alive at time of fire and then killed by the wildfire (i.e., fire-killed) would have less deep char and retain much more structural complexity.

METHODS

Study area

We conducted our field sampling across three fires that burned in 2012, 2013, and 2014 in Tweedsmuir and Entiako Provincial Parks, which are situated in the sub-boreal forests on the southern portion of the Nechako Plateau in BC (Fig. 1). The study area has a mean monthly maximum temperature of 8.5°C (range −3.3 to 19.8°C), a mean monthly nighttime temperature of −2.8°C (range −11.9 to 6.7°C), and total annual precipitation of 507.6 mm with a monthly mean of 42.3 mm (range 22.7–60.8 mm), based on the monthly means from the 1981–2010 climate normals (Abatzoglou et al. 2018). Precipitation accumulates as snow in the winter and rain during the remainder of the year. Although it rains through the summer (Abatzoglou et al. 2018), there are weeks with no rain that are associated with persistent high-pressure ridges (Nash and Johnson 1996). Within the fire perimeters, landscapes are associated with the Sub-Boreal Pine Spruce and Sub-Boreal Spruce biogeoclimatic zones (Meidinger and Pojar 1991), and lodgepole pine is the dominant canopy species (Fig. 1; BCMFLNRO 2012). Moisture gradients dictate composition, structure, and disturbance history, based on historical reconstructions from surrounding areas (Stevenson 2001, Francis et al. 2002) and stand age distributions (DeLong 1998). Within our fire perimeters, climax lodgepole pine inhabits the driest end of the moisture gradient, seral lodgepole pine persists with mean fire returns of 100–175 years, and climax spruce (*Picea engelmannii* × *glauca*) occupies pockets with high moisture levels such as riparian zones or through succession with long intervals of no fire (Parminter 1992). The landscape is gently rolling with low topographic relief, minimizing the topographic influence on fire behavior. Elevation ranges from 850 to 1300 m, in the region.

Field sampling occurred within three wildfire perimeters (Fig. 1). All fires were lightning ignited and received minimal to no suppression activities due to wilderness management

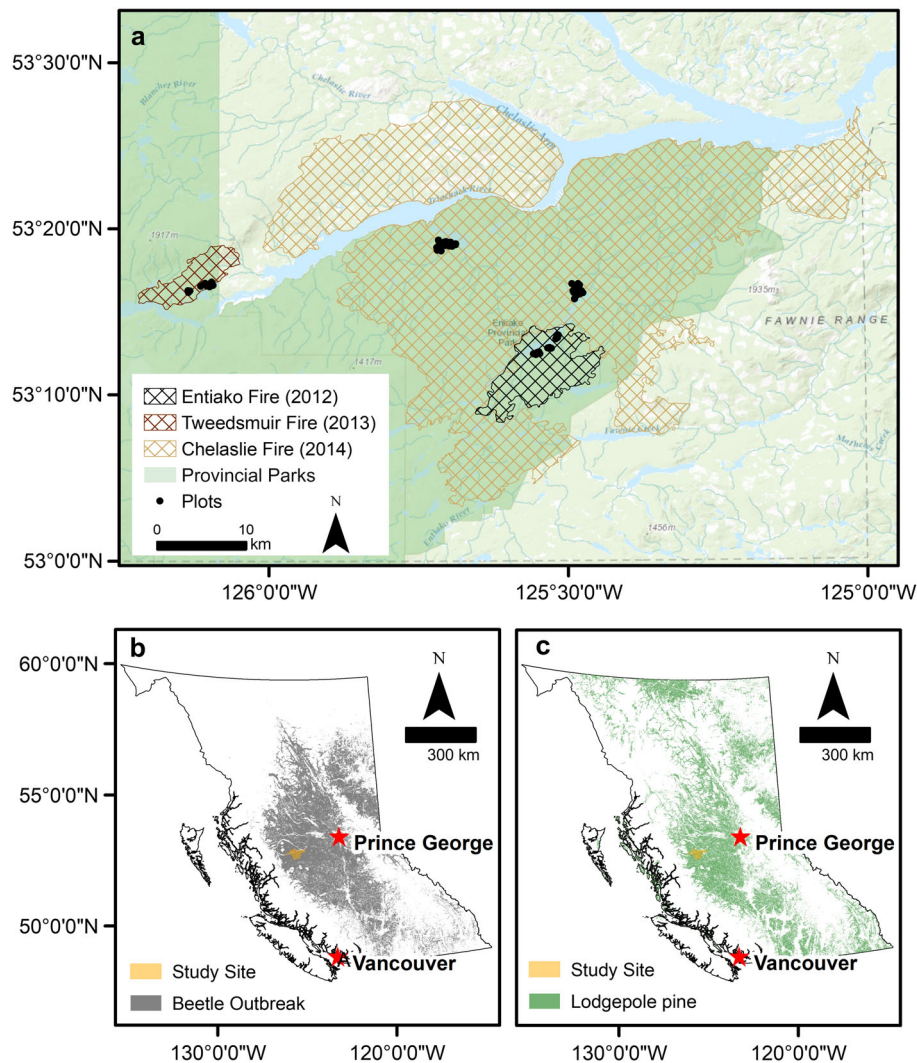


Fig. 1. Maps of the study area, MPB outbreak extent, and lodgepole pine range. Fire perimeters for three study fires that burned in 2012 (Entiako Fire; 7459 ha), 2013 (Tweedsmuir Fire; 3354 ha), and 2014 (Chelaslie Fire; 133,000 ha). (a) Provincial park boundaries are displayed as protected areas and overlaid with fire perimeters. Panel (b) shows the extent of the MPB outbreak across British Columbia based on aerial survey data from 2000 to 2011 (BCMFLNRO 2016). Panel (c) shows the estimated range of lodgepole pine across British Columbia.

objectives for the parks (Rob Krause and Mike Pritchard, *personal communication*). The Entiako Lake fire (R10171) burned 7450 ha from 3 August 2012 to 22 September 2012 (BCWS 2016). The Tweedsmuir fire (R10252) burned 3600 ha from 9 September 2013 to 16 September 2013 (BCWS 2016). The Chelaslie River fire (R10070) burned 133,100 ha from 8 July 2014 to 26 October 2014 (BCWS 2016). All fires burned through gray-phase outbreak conditions of varying

severity. The recent MPB outbreaks, circa 1990s and 2000s, peaked in the region around 2003/2004 and have affected forests across much of the province (Fig. 1). MPB activity has declined since 2006 (Wulder et al. 2009). The lag between peak outbreak and the three wildfires was about a decade, with standing dead trees (snags) beginning to transition to coarse woody debris in some outbreak-affected stands. The location of fires within parks provided a rare opportunity to

study MPB-fire interactions without the interference of active management (i.e., harvest/salvage activity, fire management/suppression).

Sampling design

Field sampling occurred from late June through August 2016, allowing us to characterize a snapshot of early successional forest communities from two to four years postfire. The study area has no road access, and sampling was limited by hiking and boating distances from three remote cabins within the parks (Fig. 1). Study plots were distributed through forest dominated by lodgepole pine. We selected plots based on a two-pronged approach including an a priori site selection using digital data, followed by verification and final selection in the field. A priori digital data included aerial survey data of MPB outbreak severity (BCMFLNRO 2016), burn severity maps generated from the differenced normalized burn ratio (dNBR; Eidenshink et al. 2007), and vegetation maps from the province's Vegetation Resource Inventory (VRI) to target areas of pure lodgepole pine (BCMFLNRO 2012). Aerial survey data for MPB outbreaks were a coarse resolution (400 m raster data products) and indicated areas within and around our study sites ranged between 50% and 100% canopy mortality (BCMFLNRO 2016). We selected from 12 to 39 plots in each study fire, depending on accessibility, for a total of 63 plots (Appendix S1: Table S1). We distributed plots across a gradient of fire severity within each fire class based on dNBR maps as low ($n = 22$), moderate ($n = 18$), and high ($n = 23$) that equated to light surface, severe surface, and crown fire based on our field measurements.

We visited plots on the ground for data collection. We verified canopy trees were predominantly lodgepole pine and were representative of the fire severity in the surrounding area. From the plot center location, we recorded GPS coordinates using a Garmin hand-held unit (GPSMAP 78s) and established a ten-by-ten-meter (100 m²; 0.01 ha) survey plot, and divided it into four quadrants along the north-south and east-west axes, identified as NE, SE, SW, and NW. Within each quadrant, we placed a one-by-one-meter (1 m²) subplot at increasing distances from the center of the plot (NE-1 m, SE-2 m, SW-3 m, and NW-4 m). Within plots, we recorded information

for each live or dead tree: species, live/dead, evidence of MPB activity including exit holes and j-shaped galleries, diameter at breast height, and measures of first-order fire effects to characterize fire severity (including legacy structures) described in detail below. Within subplots, we recorded surface fire severity as first-order fire effects including duff depth, exposed mineral soil, terrestrial surface char, and litter. Measured variables at the plot and subplot resolution were used to characterize stand structure, MPB outbreak severity, fire severity and to analyze the relationship between MPB outbreaks and fire severity.

Mortality status of canopy trees and outbreak severity

We identified trees as live or dead at time of sampling and assigned a cause of death to each dead tree. We used these data to quantify MPB outbreak severity, mortality from fire, and cumulative mortality for each plot. We identified a tree's cause of death based on protocols adapted from Harvey et al. (2013, 2014a). We attributed each tree as most likely to have been (1) killed prefire by MPB (i.e., MPB-killed), (2) killed prefire by another agent (i.e., other-killed), (3) killed by fire (i.e., fire-killed), or (4) live postfire with no evidence of MPB activity (Table 1). We evaluated snags for MPB activity unless they were alive at time of sampling. We assessed each dead canopy tree for presence or absence of exit holes associated with adult beetles emerging from the tree (Harvey et al. 2013, 2014a). Then, we removed bark from each dead tree to identify galleries specific to MPB or other bark beetle species (Harvey et al. 2013, 2014a). We classified a tree as MPB-killed if it had the requisite exit holes and j-shaped galleries specific to MPB. While much of the prefire tree mortality present was linked to MPB, we also observed significant Ips beetle (*Ips pini*) activity, which we included as other-killed if there was no evidence of MPB. We classified a tree as other-killed if it was lacking evidence of exit holes and j-shaped galleries, but other evidence suggested death prior to fire such as no needle retention in the canopy, sloughing bark, other insect activity, and decay at the base, which is common in this system due to the moist climate (Table 1); this was a small portion of the total trees sampled (7%). We classified a tree as

Table 1. The criteria and classes used to identify a tree's cause of death for the study region in BC.

Cause of death	Description	Trees sampled (%)
Live tree	Live when sampled; green canopy; no visible beetle activity	4.67
Fire-killed	Dead when sampled; scorched bark, branches, and/or outer sapwood; no evidence of galleries or exit holes from MPB or other bark beetle activity; not highly decayed/weathered particularly at the base and in the canopy	28.95
MPB-killed	Dead when sampled; no needles remaining in the canopy; vacated mountain pine beetle (MPB) galleries in cambium with exit holes in remaining bark	59.38
Other-killed	Dead when sampled; highly decayed/weathered, no bark, missing branches; more advanced decay than MPB-killed trees; full deep char with no identifiable vacated MPB galleries	7.00
Prefire-killed: MPB-killed + Other-killed	All prefire-killed from both MPB-killed and other-killed	66.38

Notes: Methods adapted from Harvey et al. (2014a). Trees sampled summarize observed data from field collections.

fire-killed if it had red needles in the canopy or postfire needle drop, and no evidence of prefire MPB or other beetle activity. We estimated a general metric of prefire-killed trees as the combination of MPB-killed and other-killed (Table 1). We calculated plot-level metrics for outbreak severity as the proportion of MPB-killed trees per plot and prefire mortality as the proportion of all prefire-killed trees per plot. Pre-outbreak stand estimates were based on all trees in the plot, regardless of status.

Fire severity recorded as first-order fire effects at the plot level

We characterized fire severity with seven measures of first-order fire effects that were scaled to a plot-level metric. We measured fire effects on standing trees/snags and the terrestrial surface including: height of scorch and/or char on trees, percent cover of scorch and/or char on trees, percent deep charring on trees, litter/duff depth, proportion of remaining litter, proportion of terrestrial surface char, and proportion of exposed mineral soil. Scorch and deep char are visually distinct, scorch with a dusty, matte black appearance and deep char with an iridescent black, scale-like appearance. In some cases, snags had both areas of scorch and deep char. The height of scorch and/or deep char (hereafter scorch/char) was measured to the nearest 0.5 m on each tree with four-meter measuring sticks and converted to mean scorch height per plot. We estimated the percent area covered, as height and circumference, of scorch and/or deep char (hereafter scorch/char) and calculated a mean proportional

area per plot. We inverted the mean proportion of area per plot to the proportion of unscorched area per plot for analysis. We recorded deep char for each tree as no deep char, <50% deep char, or 50–100% deep char coverage on the snag and calculated the proportion of snags with deep char for a plot-level variable. The four terrestrial surface fire effects metrics were measured in each subplot in the four quadrants of the plot. Litter/duff depth was measured as the combination of litter plus duff to the nearest millimeter in two opposing corners of each subplot and averaged to a plot-level variable. We recorded the percent of remaining litter, terrestrial surface char, and exposed mineral soil and calculated a mean for each variable from the four subplots to generate plot-level metric. Remaining litter, terrestrial surface char, and exposed mineral soil were converted to proportions for analysis purposes. Because we surveyed plots between two and four years postfire, we captured various early successional stages in postfire litter accumulation and vegetative regrowth.

Fire severity recorded as biomass consumption of legacy structure at the tree level

To characterize the effect of outbreaks and wildfires on postfire legacy structure, we categorized biomass loss on each tree based on the remaining branch structure and deep char. The remaining branch structure refers to the fine, moderate, and coarse branch structure, and it was quantified as presence or absence. A classification of absence meant that there was no remaining branch structure on the tree and no

associated branches on the ground in the area of the tree/snag, which indicated that branches were consumed by fire. As described above, deep char was visually distinct from scorch and recorded as absent, <50%, or 50–100% deep char coverage on each snag. We retained these categories to assess the relationship between deep char and remaining branch structure and converted the categories to presence or absence of deep char for each tree/snag to evaluate the relationship between a tree's cause of death and deep char development.

Fire weather and topography

The Entiako, Tweedsmuir, and Chelaslie fires that provided the footprint for our study burned during three different fire seasons (2012, 2013, and 2014) across a landscape with low topographic complexity. Fires burned over a relatively long duration within each season, which allowed us to account for variability in fire weather and day-of-burn conditions (Appendix S1: Table S1). Each plot was assigned a day of burn from day-of-burn progression maps estimated from MODIS hotspot data (Parks 2014), which allowed us to assign the daily fire weather index (FWI) to each plot that was generated from the nearest weather station (Appendix S1: Table S1). The calculated FWI is a metric from the Canadian Forest Fire Weather Index System (Van Wagner 1987) that integrates temperature, relative humidity, and wind speed. We used the FWI to assign each plot a burning condition category of moderate (>13–29) or extreme (>29), based on breakpoints outlined by Alexander and De Groot (1988). All fires experienced moderate burning conditions within a portion of their perimeter; however, extreme burning conditions ($\text{FWI} \geq 29$) only occurred in two of the three fires (2012 Entiako and 2014 Chelaslie fires; Appendix S1: Table S1). For plots, elevation fluctuated between 873 and 1043 m. Plots were relatively flat with a mean slope of 2.6° (range $0\text{--}20^\circ$). We did not pursue topography as an explanatory variable of fire severity due to the low topographic variability at our study plots.

Statistical analysis

We tested for relationships between each of the seven fire effects metrics and MPB outbreak

severity at the plot level, while accounting for burning conditions. Our seven fire effects metrics served as response variables: average scorch/char height, average proportion of unscorched/uncharred area on trees, proportion of trees with deep char, litter/duff depth, proportion of remaining litter, proportion of terrestrial surface char, and proportion of exposed mineral soil. We tested the relationship of each response variable against the proportion of MPB-killed trees (our index of MPB severity) and burning conditions, which was included as a categorical variable of moderate or extreme FWI (Appendix S1: Table S1). An interaction term between burning conditions and proportion of MPB-killed trees was included in all models to assess whether observed relationships changed under different fire weather conditions. Relationships with scorch/char height and litter/duff depth were fit using linear models. The proportion of terrestrial surface char was logit-transformed and fit with a linear model. All other models in which the response variable was a proportion were analyzed with generalized linear models, and each response variable was fit with a distribution appropriate for the type and distribution of the response variable (see Appendix S1: Table S3 for distributions associated with each analysis). We also ran each model and replaced the proportion of MPB-killed trees with the proportion of pre-fire-killed trees, since dead trees would all be similar in terms of conditions and moisture content regardless of what killed them. The models with the proportion of prefire-killed trees demonstrated similar statistical relations to the proportion of MPB-killed trees. We report all models that were statistically significant, and we kept all fire effects models that were run with the proportion of MPB-killed trees as an explanatory variable, since these models were a more conservative estimate of MPB caused mortality.

We evaluated the effect of outbreak severity and wildfire on postfire legacy structure from tree-level fire effects of deep char and branch structure loss. We analyzed data at the tree level using two different response variables: (1) presence/absence (1/0) of branch structure on individual trees and (2) presence/absence (1/0) of deep char on individual trees. We accounted for burning conditions as a categorical variable of moderate or extreme FWI (Appendix S1: Table S1). An

interaction term between burning conditions and cause of death was included in all models. We used generalized linear mixed models with a binomial distribution for presence/absence data using a logit link, and each model included the plot as a random effect and the interaction term between cause of death and burning conditions. Results are reported as probability of occurrence, and the comparison between mortality types (e.g., MPB-killed versus fire-killed) is reported as the odds ratio. Additionally, we determined whether the presence/absence of branch structure was related to the coverage of deep char on the tree. Our explanatory variable of deep char was treated as a three-level categorical variable of no deep char, <50% coverage of deep char, or 50–100% coverage of deep char on the tree, while accounting for burning conditions.

We assessed fit for all models by visually inspecting the residuals, which appeared to be adequately met. We evaluated and corrected for overdispersion in all generalized linear models and generalized linear mixed models when necessary. For our two linear models, assumptions of normality and constant variance of the residuals were checked graphically and appeared to be adequately met. We assessed the interaction term with a drop-in-deviance test. The interaction term was retained in each model regardless of statistical significance, because of the known interaction between fire weather and fuels. All statistical analyses were conducted in R statistical computing software version 3.4.4 with the stats package (R Development Team 2018). For generalized linear models, we used the function `glm` in the MASS package (Venables and Ripley 2002). For generalized linear mixed models, we used the function `glmer` in the lme4 package (Bates et al. 2015). We considered $p < 0.05$ as convincing evidence of a relationship and $P < 0.10$ as suggestive of a relationship to minimize the potential of a type II error. Data and code for analyses are available online (Talucci 2019).

RESULTS

We collected data from 943 trees across 63 field plots with 910 lodgepole pine trees/snags and 33 spruce trees/snags. Canopy tree species were predominantly lodgepole pine with a plot mean of

96% (range across plots 63–100%; Appendix S1: Table S2). Estimated mortality from MPB was 59% of all trees sampled, and estimated prefire mortality (i.e., MPB-killed plus other-killed) was 66% of all trees sampled (Table 1). When we evaluated just lodgepole pine mortality across all 63 plots, the estimated mean for lodgepole pine killed by MPB was 63% and the estimated mean for lodgepole pine killed by all agents prior to fire (all prefire) was 70% (Appendix S1: Table S2). Cumulative mortality was estimated at 93% for lodgepole pine as a combination of prefire and fire mortality (Appendix S1: Table S2).

Effect of outbreak severity on first-order fire effects at the plot level

The effect of outbreak severity on fire severity was limited, with six of seven fire effects showing no evidence of an effect (Fig. 2, Appendix S1: Table S3). average scorch/char height, average proportion of unscorched/uncharred area on trees, litter/duff depth, proportion of remaining litter, proportion of terrestrial surface char, and proportion of exposed mineral soil showed no evidence of an effect of outbreak severity (Fig. 2, Appendix S1: Table S3). Outbreak severity did show evidence of an effect on proportion of trees with deep char. Under moderate burning conditions, the proportion of trees with deep char increased with increasing outbreak severity (Fig. 2, Appendix S1: Table S3), which held true when we substituted the proportion of prefire-killed trees for MPB-killed trees (Fig. 2, Appendix S1: Table S3). Under extreme burning conditions, the relationship between the proportion of MPB-killed trees and deep char was not statistically significant, however when we substituted the proportion of prefire-killed trees for MPB-killed trees that relationship was statistically significant (Fig. 2, Appendix S1: Table S3).

Effect of outbreak and wildfire on legacy structure

Outbreak severity and wildfire showed distinct evidence of an effect on the legacy structure of the forest, measured by biomass consumption as deep char and branch structure loss on individual trees. Both deep char development and branch structure loss had greater odds of occurrence when a tree was dead prior to fire (i.e., MPB-killed or prefire-killed) compared to being

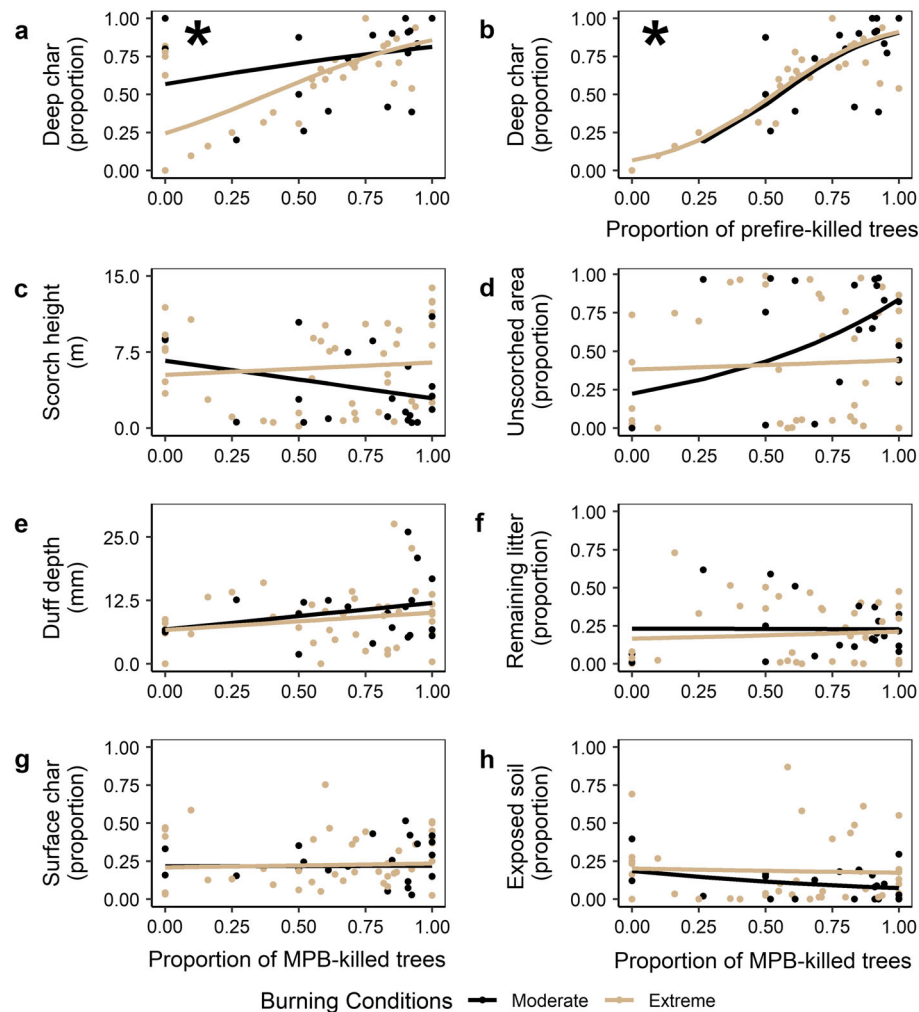


Fig. 2. The relationships between outbreak severity and seven first-order fire effects measured at the plot level: proportion of trees with deep char (a–b), average scorch/char height (c), average proportion of unscorched/un-charred area on trees (d), litter/duff depth (e), proportion of remaining litter (f), proportion of terrestrial surface char (g), and proportion of exposed mineral soil (h). Response variables are along the *y*-axis with the explanatory variable of the proportion of mountain pine beetle (MPB) killed trees or prefire-killed trees (only panel b) along the *x*-axis. Points are the raw data ($n = 63$ plots), and fitted lines show the estimated statistical relationship. The response variable of deep char is shown in panels a and b, and they were the only two models that indicated a strong statistical relationship (*). Response variables c–h were unrelated to outbreak severity. See Appendix S1: Table S3 for model estimates and confidence intervals.

alive at time of fire, which was consistent across both moderate and extreme burning conditions (Figs. 3, 4, Appendix S1: Table S4). Under both moderate and extreme burning conditions, a MPB-killed and prefire-killed snag had greater odds of developing deep char compared to a fire-killed tree (Figs. 3, 4, Appendix S1: Table S4). There were greater odds of branch structure

being consumed on a MPB-killed and prefire-killed snag compared to a fire-killed tree under moderate conditions, and the size of that effect was slightly smaller under extreme conditions but still significant (Figs. 3, 4, Appendix S1: Table S4). We found that branch structure had greater odds of being consumed when deep char exceeded 50% coverage on the tree for both

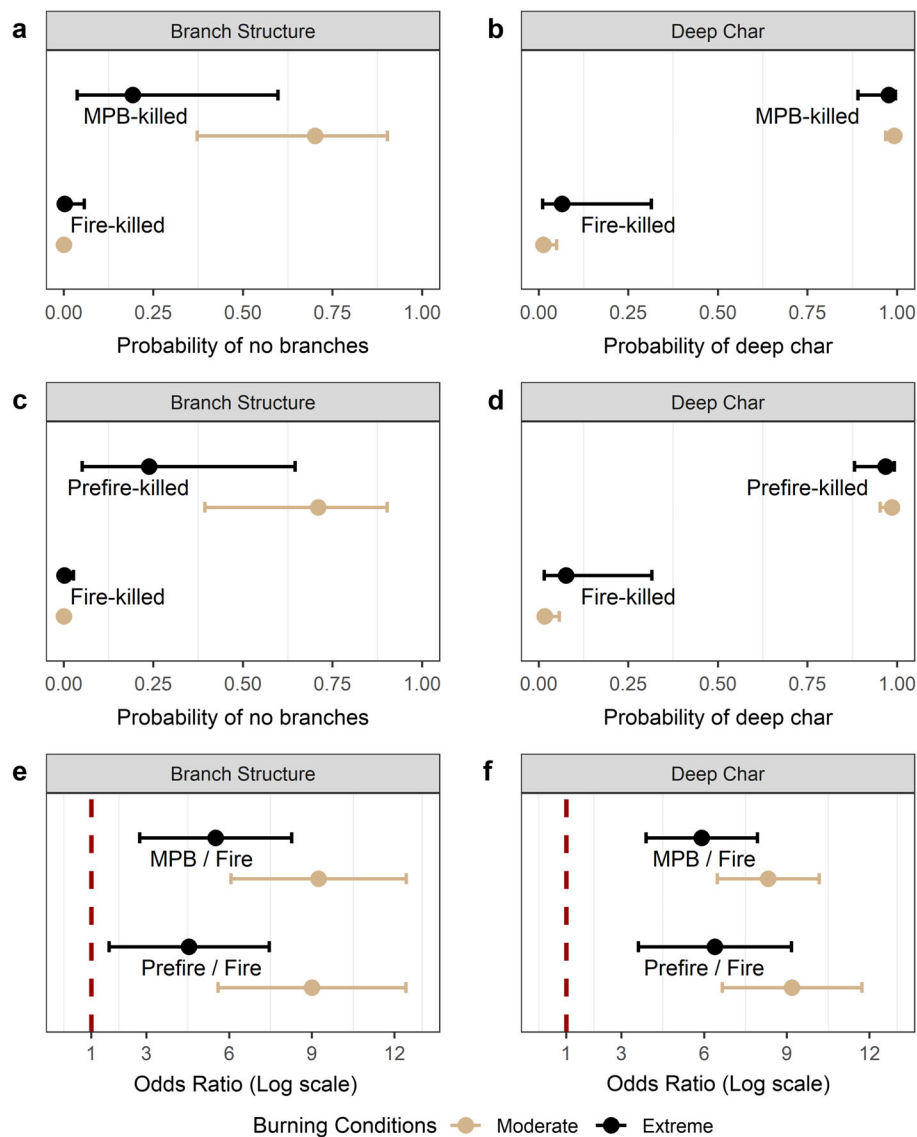


Fig. 3. The effect of outbreak severity and wildfire on legacy structure measured at the tree level. Tree-level fire effects show the role of mountain pine beetle (MPB) outbreak severity (MPB-killed and prefire-killed) on consumption of woody material and simplification of structural legacies in the form of branch loss and deep char development (a–d). Comparison between groups (i.e., MPB-killed versus fire-killed) is shown in (e) and (f) as odds ratios with the red dashed line marking no difference at one. The model estimates are listed in Appendix S1: Table S4.

moderate and extreme burning conditions (Fig. 5, Appendix S1: Table S5).

DISCUSSION

We found that fire severity as measured by scorch/char height and area, and surface fire

metrics, is not influenced by MPB outbreak severity but that fire severity measured as biomass loss and legacy structure was consistently influenced by the outbreak history. These findings from BC align with previous field research that evaluated the influence of outbreak severity on fire severity in the western United States

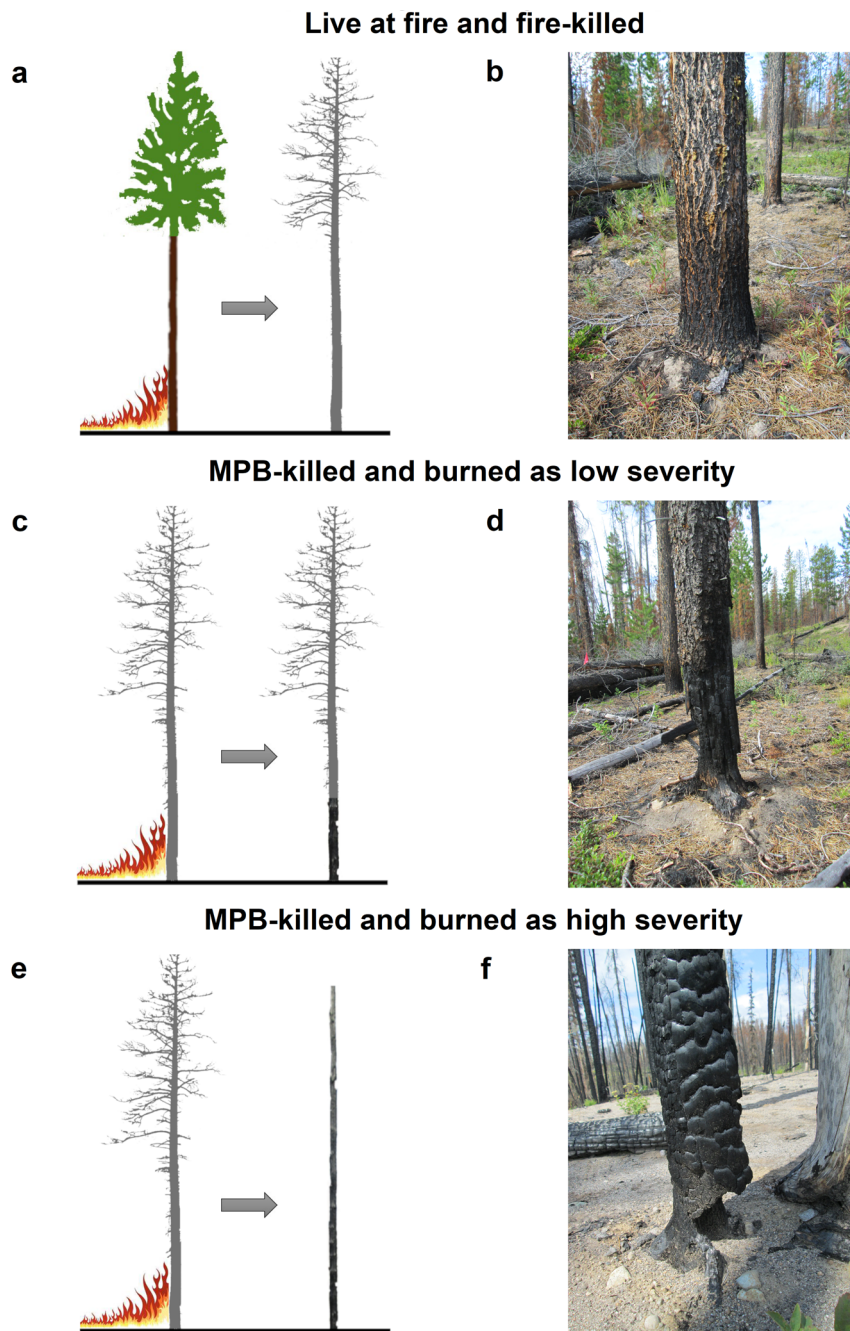


Fig. 4. Tree-level fire effects are dependent on whether a tree is alive or dead at time of fire. Panel (a) illustrates a tree that is live at time of fire and killed by fire, with the adjacent panel (b) showing a photo of a tree live at time of fire and killed by fire with scorched bark but no consumption of the tree. Panel (c) illustrates a MPB-killed tree that burns under low severity conditions, with the adjacent panel (d) showing a photo of deep char development and consumption at the base of the tree, which is attributed to fungal development (Donato et al. 2009). Panel (e) illustrates a MPB-killed tree that burns under high-severity conditions, with the adjacent panel (f) showing a photo of deep char that covers the entire tree in a plot that burned as high severity.

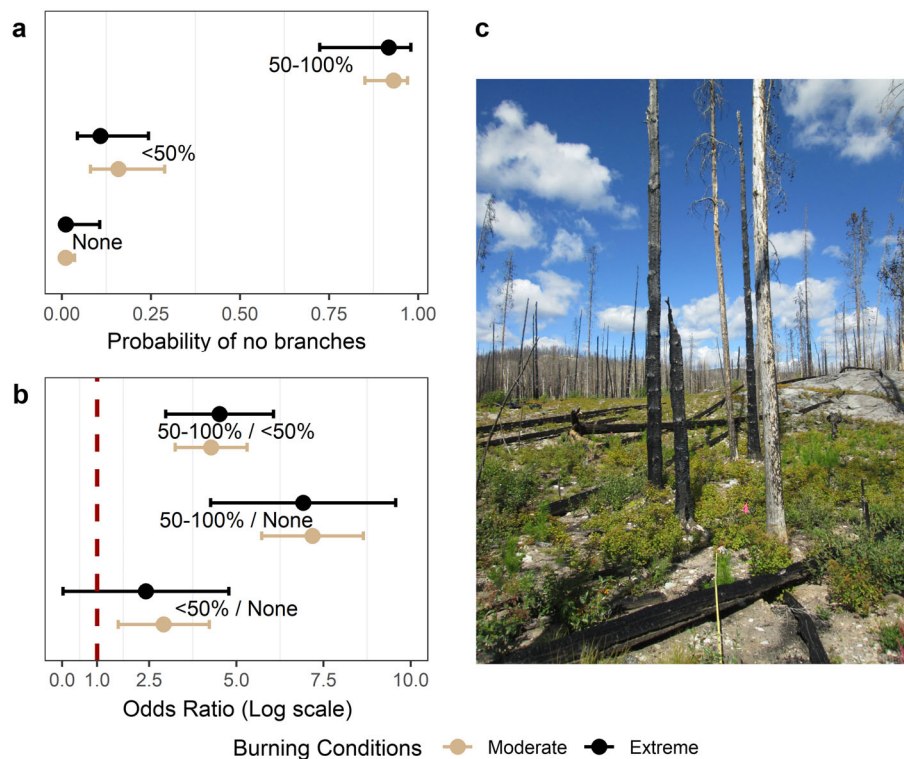


Fig. 5. Deep char coverage influences the consumption of branch structure on a tree. The probability of branch loss from deep char development is shown in (a). Comparisons between groups are shown in (b) with a red dashed line demarcating no difference at one. In (c), a photo of a snag with deep char and no branches adjacent to a snag with branches still intact and no deep char. Model estimates are listed in Appendix S1: Table S5.

(Harvey et al. 2014a, b, Agne et al. 2016), but extends our understanding of these short-interval disturbances by highlighting the synergistic effect on postfire structural legacies. Prefire mortality had a greater likelihood for increased biomass consumption and deep char, which aligns with findings on reburns, where wildfires recur in short intervals (Donato et al. 2016). Prefire mortality, regardless of the mechanism of death, results in an altered legacy structure that is more simplistic and has more deep char. While this effect on structural legacies is generally accepted in the field, it has been broadly overlooked and unquantified in assessments of how outbreak affects fire severity. When asking the question “does MPB outbreak affect fire severity, are they linked disturbances,” the answer is yes—specifically through the deadwood structure that remains in these ecosystems.

Effect of outbreak severity on first-order fire effects at the plot level

Outbreak severity did not show evidence of an effect on fire severity for six out of seven measured first-order fire effects; the exception was deep char. This reflects similar findings in gray-phase outbreak conditions found by Harvey et al. (2014b) and extends our understanding to the geography of BC’s sub-boreal forests. Our finding indicates some inherent noise and uncertainty in our data as well as the influence of fire weather. The six fire effects—scorch/char height and area, duff depth, litter, surface char, and exposed soil, are likely controlled by the combined factors of the fire environment, that is, the interaction of fuels, weather, and topography (Countryman 1972, Krawchuk and Moritz 2011, Whitman et al. 2015) but without a strong signal from outbreak fuel structure, which aligns

with previous research evaluating interactions between outbreak severity and fire severity (Harvey et al. 2014a, b, Agne et al. 2016). Scorch on trees is naturally variable and can be driven by multiple factors including the composition of fuel structures, crown and/or surface fire spread, burning conditions, slope steepness, and ignition patterns (Alexander and Cruz 2012a). Our results show no evidence of a relationship between terrestrial surface fire effects and outbreak severity, which was also consistent with findings in previous retrospective studies with gray-phase outbreak conditions (Harvey et al. 2014b, Agne et al. 2016). The lag time between needle drop and our study fires would have allowed for the decomposition of fine fuels (Simard et al. 2011, Harvey et al. 2013) thus minimizing the effect of outbreak on surface fuels. Most snags were still standing at time of fire, so that the concern of increased surface fire severity from abundant coarse woody debris was not observed. These findings support the general narrative that low-frequency and high-severity fire regimes associated with lodgepole pine in sub-boreal forests are strongly driven by climate systems of high-pressure, creating dry-hot conditions conducive for burning such that variability in fuel structure/vegetation plays a secondary role (Bessie and Johnson 1995, Nash and Johnson 1996, Whitman et al. 2015).

Effect of outbreak and wildfire on structural legacies

Our findings support the notion that dead wood, which in our landscapes is predominantly snags generated by MPB outbreaks, burns differently than live wood and indicates an important MPB-fire connection. Live trees rarely experience significant combustion, and therefore, little to no consumption occurs on the tree (Campbell et al. 2007), which is attributed to higher moisture content compared to their dead counterparts (Brown et al. 1985). Extended periods of smoldering and glowing combustion (Brown et al. 1985, Page and Jenkins 2007, Hyde et al. 2011) are facilitated by lower moisture content in snags and coarse wood (Stockstad 1979). Lower moisture content in snags could enable passive crown fire or torching of snags (Wenger 1984), which may be the primary mechanism for consumption of branch structure. Some simplification of branch

structure may also occur on gray-phase MPB-killed trees prior to fire. The torching of snags and extended periods of smoldering have been demonstrated in areas that experience reburn, wildfires that recur in short intervals (Donato et al. 2016). High-severity reburns have shown there is an eightfold increase in deep char development on snags and the retention of woody biomass is half the amount of once burned areas, in the Klamath Mountains of southwestern Oregon (Donato et al. 2016). Where wind-throw is followed by wildfire in short intervals, snags and coarse wood have been shown to be reduced with a marginal increase in charred material (Buma et al. 2014). In lodgepole pine/Douglas-fir (*Pseudotsuga menziesii*) forests on the Chilcotin Plateau of BC south of our study sites, areas of high prefire mortality from MPB outbreak experienced 13% more consumption of dead wood and the variability in canopy consumption was attributed to mortality status with dead prefire snags having more of their branch structure consumed (Brad Hawkes, *personal communication*). This evidence indicates that it is not necessarily the mechanism of prefire mortality, for example, MPB outbreak, wind-throw, or prior wildfire, but the fact that there is an abundance of deadwood with altered moisture levels and fuel structures compared to live wood, which alters postfire ecological and structural legacies as they relate to standing snags and coarse woody debris.

The consumption of branches and deep char development on snags alters the structural legacies that endure through fire. These altered legacies may introduce long-term implications for ecosystem structure and function including availability of canopy seedbank, accumulation of coarse woody debris, and early seral structure and resources for early seral species (Franklin et al. 2000, Swanson et al. 2011, Johnstone et al. 2016). After MPB outbreak, lodgepole pine snags continue to retain some of their aerial seedbank in the canopy postmortem while some cones fall to the forest floor (Teste et al. 2011). Cones in snags or on the forest floor can be exposed to extended heating from a snag smoldering or slower moving surface fire (Alexander and Cruz 2012b), which could reduce seedbanks and influence postfire resilience (Johnstone et al. 2016). The loss in snag biomass and branch structure alters the accumulation of coarse wood that may

influence short- and long-term carbon and nutrient cycles (Harmon 2001), structure of habitat for wildlife (Fontaine et al. 2009, House 2014) including nesting and perching habitat, and both structure and function of early seral ecosystems (Swanson et al. 2011). More charring on trees reduces the quality of the snag for saproxylic insects thereby affecting foraging woodpeckers (Saint-Germain et al. 2004, Nappi et al. 2010), which could influence trophic webs. Deep char development can encapsulate the remaining wood, which may limit decomposition, slow decay, and extend long-term carbon storage (Preston 2009, Bird et al. 2015). Together, these changes to dead wood that may alter the long-term structure and function in the postfire forest are considered compound disturbance effects, where the outbreak severity and fire severity work in combination to create unique post-disturbance conditions that are different than the outcomes of the singular disturbance of wildfire (Paine et al. 1998). Further research is needed to determine the long-term implications of compound disturbance effects related to legacy structure, coarse wood recruitment, carbon storage, pyrogenic carbon, habitat structures, trophic webs, and early seral ecosystems in forests where fires are increasingly burning through stands with high volumes of snags from insects, windthrow, drought, and prior fire.

CONCLUSION

Sub-boreal forest ecosystems of BC have experienced widespread tree mortality from the MPB outbreaks, generating a fuel structure characterized by an abundance of deadwood that is now interacting with wildfires. The contiguous landscape of lodgepole pine-dominated forests situated at the epicenter of the outbreak in western North America allowed us to assess interacting, or linked, effects between outbreak and fire severity. Our results suggest that while many first-order fire effects are not influenced by outbreak severity, legacy structure related to the degree of biomass consumption is strongly influenced by the interaction of outbreak severity and fire severity. These findings are especially important to consider after the 2017 and 2018 fire seasons in BC wherein a record number of hectares

burned, with many of the fires burning through snag forests affected by MPB outbreaks.

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